



2021 Caringbah EXT2 AT3 Exam

Maths extension 2 (Sydney Technical High School)



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Caringbah High School

Year 12 2021

Mathematics Extension 2

HSC Course

Assessment Task 3

General Instructions

- Reading time – 5 minutes
- Working time – 70 minutes
- Write using a black pen
- Board-approved calculators may be used
- A reference sheet is provided.
- In Questions 6-9, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for partial or incomplete answers

Total marks – 45

Section I 5 marks

Attempt Questions 1-5
Mark your answers on the answer sheet provided. You may detach the sheet and write your name on it.

Section II 40 marks

Attempt Questions 6-9
Write your answers in the answer booklets provided. Ensure your name or student number is clearly visible.^[SL1]

Name: _____

Class: _____

Marker's Use Only						
Section I	Section II				Total	
Q 1-5	Q6	Q7	Q8	Q9		
/5	/10	/10	/10	/10	/45	%

Section I

5 marks

Attempt Questions 1-5

Allow about 8 minutes for this section

Use the multiple-choice answer sheet for Questions 1–5

1. Which expression is equal to $\int \tan x \, dx$?

(A) $\sec^2 x + C$

(B) $-\ln |\cos x| + C$

(C) $\frac{\tan^2 x}{2} + C$

(D) $\ln |\sec x + \tan x| + C$

2. Which of the following is an expression of $\int x^3 e^{x^4} \, dx$?

(A) $x^3 e^{x^4} + C$

(B) $e^{x^4} + C$

(C) $4e^{x^4} + C$

(D) $\frac{1}{4}e^{x^4} + C$

3. The diagram shows a cube of side 2 units.

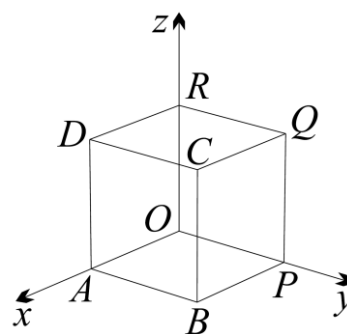
What is the length of the diagonal OC ?

(A) $\sqrt{6}$

(B) $2\sqrt{2}$

(C) $2\sqrt{3}$

(D) $3\sqrt{2}$



4. Which of the following is the first line of working for the expression:

$$\int x \cos x \, dx$$

(A) $-x \sin x - \int \cos x \, dx$

(B) $-x \cos x - \int \sin x \, dx$

(C) $x \sin x - \int \cos x \, dx$

(D) $x \cos x - \int \sin x \, dx$

5. Find any values of λ for which $\underline{a}$ and $\underline{b}$ are perpendicular if

$$\underline{a} = \begin{bmatrix} 1 + \lambda \\ 2\lambda \\ \lambda + 2 \end{bmatrix} \text{ and } \underline{b} = \begin{bmatrix} 2 \\ 5 - 4\lambda \\ 2\lambda - 4 \end{bmatrix}$$

(A) $\lambda = 0$

(B) $\lambda = -1$

(C) $\lambda = 1$

(D) λ has no solution

End of Section I

Section II

60 marks

Attempt Questions 6–9

Allow about 1 hour for this section

Answer each question in a SEPARATE writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (10 marks) Use a SEPARATE writing booklet.

- (a) The points P and Q have position vectors $1\hat{i} - 2\hat{j} - 3\hat{k}$ and $4\hat{i} - 5\hat{j} - 6\hat{k}$ respectively. Find

(i) $\overrightarrow{PQ}$ 1

(ii) $\overrightarrow{QP} - 2\overrightarrow{PQ}$ 1

(iii) $|\overrightarrow{PQ}|$ 1

- (b) Evaluate 2

$$\int_0^{\frac{\pi}{3}} \cos^3 \theta \, d\theta$$

- (c) Find the Cartesian equation of the line 2

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix} + \lambda \begin{bmatrix} -1 \\ -4 \end{bmatrix}$$

- (d) Use the substitution $u = \sqrt{x}$, or otherwise, evaluate 3

$$\int_0^1 \frac{1}{(1+x)\sqrt{x}} \, dx$$

End of Question 6

Question 7 (10 marks) Use a SEPARATE writing booklet.

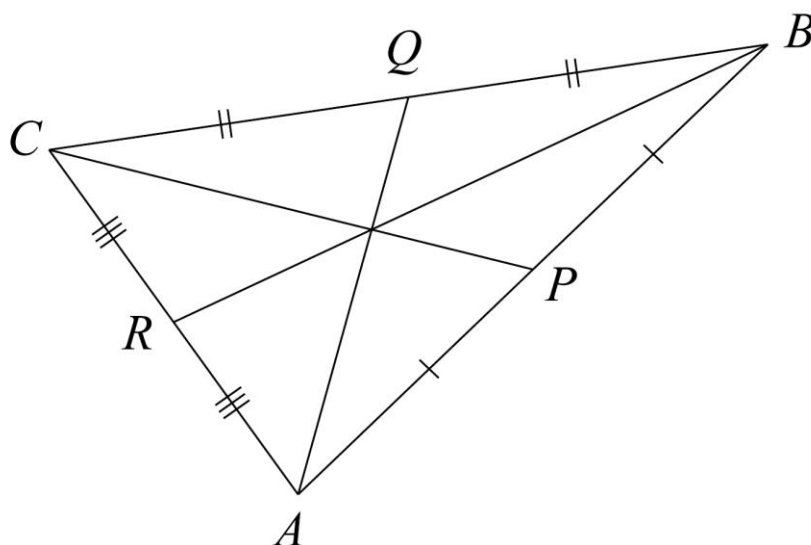
(a) Find

$$\int x \tan^{-1} x \, dx$$

3
[SL2]

(b) The diagram above shows $\triangle ABC$, where A , B and C have position vectors $\underline{a}$, $\underline{b}$ and $\underline{c}$ respectively. The points P , Q and R bisect the intervals AB , BC and CA respectively.

3
[SL3]



(i) Show that $\overrightarrow{AQ} = \frac{1}{2}(\underline{c} + \underline{b}) - \underline{a}$ **1**

(ii) Hence show that $\overrightarrow{AQ} + \overrightarrow{BR} + \overrightarrow{CP} = 0$ **2**

(c) Let $I_n = \int_0^{\frac{\pi}{2}} x^n \sin x \, dx$, for intergeters $n \geq 0$

(i) Show that $I_n = n \left(\frac{\pi}{2}\right)^{n-1} - n(n-1) I_{n-2}$ for $n \geq 2$ **2**

(ii) Hence evaluate $I_2 = \int_0^{\frac{\pi}{2}} x^2 \sin x \, dx$ **2**

End of Question 7

Question 8 (10 marks) Use a SEPARATE writing booklet.

(a) Determine $\int \sqrt{\frac{2-x}{2+x}} dx$ **3**

(b) Given $u = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$ and $v = \begin{bmatrix} -7 \\ 6 \\ -5 \end{bmatrix}$

(i) What is the scalar product of the two vectors **1**

(ii) Hence find the projection of u onto v **2**

(c) Using the substitution $t = \tan \frac{\theta}{2}$ find **4**

$$\int_0^{\frac{\pi}{2}} \frac{1}{4 \sin \theta - 2 \cos \theta + 6} d\theta$$

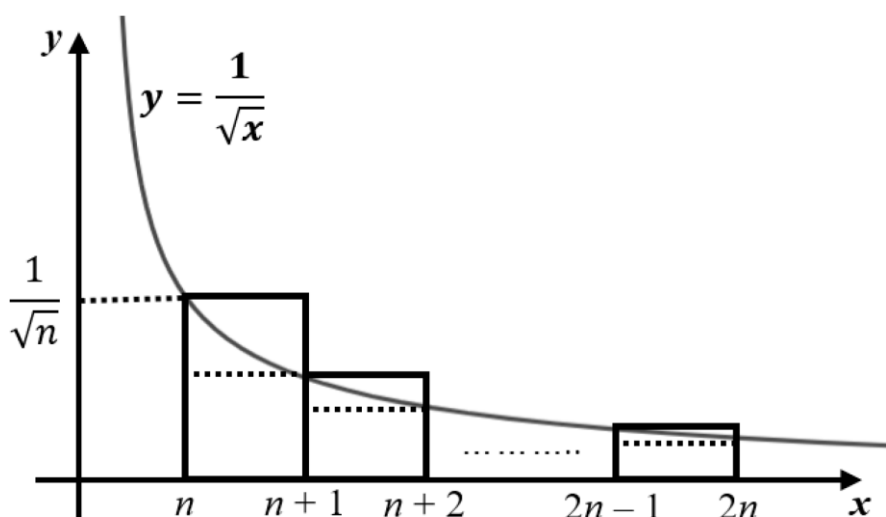
End of Question 8

Question 9 (10 marks) Use a SEPARATE writing booklet.

- (a) Find the exact values of x such that the angle between the vectors $\underline{a}$ and $\underline{b}$ is 60° if: 3

$$\underline{a} = \begin{bmatrix} 1 \\ 2 \\ x \end{bmatrix} \text{ and } \underline{b} = \begin{bmatrix} x \\ 2 \\ 1 \end{bmatrix}$$

- (b) In the diagram below, the graph of $y = \frac{1}{\sqrt{x}}$ has been drawn, and n rectangles have been constructed between $x = n$ and $x = 2n$, each of width 1 unit



- i) Show that $\int_n^{2n} \frac{1}{\sqrt{x}} dx = 2\sqrt{n}(\sqrt{2} - 1)$ 1

- ii) Let $S_n = \frac{1}{\sqrt{n+1}} + \frac{1}{\sqrt{n+2}} + \dots + \frac{1}{\sqrt{2n}}$, show that 4

$$2\sqrt{n}(\sqrt{2} - 1) + \frac{1-\sqrt{2}}{\sqrt{2n}} < S_n < 2\sqrt{n}(\sqrt{2} - 1)$$

- iii) Hence find, correct to three decimal places

$$\frac{1}{\sqrt{1\,000\,000}} + \frac{1}{\sqrt{1\,000\,001}} + \frac{1}{\sqrt{1\,000\,002}} + \dots + \frac{1}{\sqrt{2\,000\,000}}$$
 2

End of Paper

Extra Writing Space

Please clearly indicate question number.

This image shows a full page of white paper with horizontal dotted lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Multiple Choice Answer Sheet

Name _____

Colour the oval A, B, C or D that best answers the question.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct ↗

1. A ☐ B ☐ C ☐ D ☐

2. A ☐ B ☐ C ☐ D ☐

3. A ☐ B ☐ C ☐ D ☐

4. A ☐ B ☐ C ☐ D ☐

5. A ☐ B ☐ C ☐ D ☐

2021 EXT 2 AT3 SOLUTIONS

Q1 $\int \tan x \, dx = \int \frac{\sin x}{\cos x} \, dx = -\ln |\cos x| + C$ **B**

Q2 $\int x^3 e^{x^4} \, dx = \frac{1}{4} \int 4x^3 e^{x^4} \, dx = \frac{1}{4} e^{x^4} + C$ **D**

Q3 $|OC| = \sqrt{2^2 + 2^2 + 2^2} = \sqrt{12} = 2\sqrt{3}$ **C**

Q4 $u = x \quad v' = \cos x \quad \therefore u' = 1 \quad v = \sin x$
 $\therefore \int x \cos x \, dx = x \sin x - \int 1 \cos x \, dx$ **C**

Q5 $a = \begin{bmatrix} 1 + \lambda \\ 2\lambda \\ \lambda + 2 \end{bmatrix}$ and $b = \begin{bmatrix} 2 \\ 5 - 4\lambda \\ 2\lambda - 4 \end{bmatrix}$ and $a \perp b$
 $\therefore a \cdot b = 2 + 2\lambda + 10\lambda - 8\lambda^2 + 2\lambda^2 - 8 = 0$
 $\therefore \lambda^2 - 2\lambda + 1 = (\lambda - 1)^2 = 0 \therefore \lambda = 1$ **C**

Q6a) i) $\overrightarrow{PQ} = \begin{bmatrix} 4 - 1 \\ -5 - (-2) \\ -6 - (-3) \end{bmatrix} = \begin{bmatrix} 3 \\ -3 \\ -3 \end{bmatrix}$
 ii) $\overrightarrow{QP} - 2\overrightarrow{PQ} = \begin{bmatrix} -3 \\ 3 \\ 3 \end{bmatrix} - 2 \begin{bmatrix} 3 \\ -3 \\ -3 \end{bmatrix} = \begin{bmatrix} -9 \\ 9 \\ 9 \end{bmatrix}$
 iii) $|\overrightarrow{PQ}| = \sqrt{3^2 + (-3)^2 + (-3)^2} = \sqrt{27}$

Q6b) $\int_0^{\frac{\pi}{3}} \cos^3 \theta \, d\theta = \int_0^{\frac{\pi}{3}} \cos^2 \theta \cos \theta \, d\theta$
 $= \int_0^{\frac{\pi}{3}} (1 - \sin^2 \theta) \cos \theta \, d\theta$
 $= \int_0^{\frac{\pi}{3}} \cos \theta - \sin^2 \theta \cos \theta \, d\theta$
 $= \sin \theta - \frac{1}{3} \sin^3 \theta \Big|_0^{\frac{\pi}{3}} = \frac{\sqrt{3}}{2} - \frac{1}{3} \times \left(\frac{\sqrt{3}}{2}\right)^3 = \frac{3\sqrt{3}}{8}$

Q6c) $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix} + \lambda \begin{bmatrix} -1 \\ -4 \end{bmatrix} \therefore x = 3 - \lambda \quad y = 5 - 4\lambda$
 $\therefore y = 5 - 4(3 - x) \therefore y = 4x - 7$

Q6d) $u = \sqrt{x} \therefore du = \frac{1}{2\sqrt{x}} \, dx$
 $\therefore \int_0^1 \frac{1}{(1+x)\sqrt{x}} \, dx = 2 \int \frac{1}{1+u^2} \, du = 2 \tan^{-1} u$
 $= 2 \tan^{-1} \sqrt{x} \Big|_0^1 = \frac{\pi}{2}$

Q7a) Let $u = \tan^{-1} x \quad v' = x$

$$\therefore u' = \frac{1}{1+x^2} \quad v = \frac{x^2}{2}$$

$$\therefore \int x \tan^{-1} x \, dx = \frac{x^2}{2} \tan^{-1} x - \int \frac{x^2}{2} \times \frac{1}{1+x^2} \, dx$$

$$= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \frac{1+x^2-1}{1+x^2} \, dx$$

$$= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int 1 - \frac{1}{1+x^2} \, dx$$

$$= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} [x - \tan^{-1} x] + C$$

Q7b) i) $\overrightarrow{AQ} = \overrightarrow{AC} + \frac{1}{2} \overrightarrow{CB} = (c - a) + \frac{1}{2}(b - c)$
 $= c - a + \frac{1}{2}b - \frac{1}{2}c = \frac{1}{2}c - a + \frac{1}{2}b$
 $= \frac{1}{2}(c + b) - a$

ii) As $\overrightarrow{AQ} = \frac{1}{2}(c + b) - a$

Then $\overrightarrow{BR} = \frac{1}{2}(a + c) - b$ and $\overrightarrow{CP} = \frac{1}{2}(a + b) - c$
 $\therefore \overrightarrow{AQ} + \overrightarrow{BR} + \overrightarrow{CP} =$

$$\frac{1}{2}(c + b) - a + \frac{1}{2}(a + c) - b + \frac{1}{2}(a + b) - c = 0$$

Q7c) i) $I_n = \int_0^{\frac{\pi}{2}} x^n \sin x \, dx$ Let $u = x^n$; $v' = \sin x$
 $\therefore u' = nx^{n-1} \quad v = -\cos x$

$$\therefore I_n = [-x^n \cos x]_0^{\frac{\pi}{2}} + n \int_0^{\frac{\pi}{2}} x^{n-1} \cos x \, dx$$

Let $u = x^{n-1}$; $v' = \cos x$

$$\therefore u' = (n-1)x^{n-2}; v = \sin x$$

$$\therefore I_n = n \left([x^{n-1} \sin x]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} (n-1)x^{n-2} \sin x \, dx \right)$$

$$= n \left(\frac{\pi}{2} \right)^{n-1} - n(n-1) \int_0^{\frac{\pi}{2}} x^{n-2} \sin x \, dx$$

$$= n \left(\frac{\pi}{2} \right)^{n-1} - n(n-1) I_{n-2}$$

$$\text{Q7c) ii) } \int_0^{\frac{\pi}{2}} x^2 \sin x \, dx = 2 \left(\frac{\pi}{2} \right) - 2 \times I_0$$

$$I_0 = \int_0^{\frac{\pi}{2}} \sin x \, dx = [-\cos x]_0^{\frac{\pi}{2}} = 1$$

$$\therefore \int_0^{\frac{\pi}{2}} x^2 \sin x \, dx = \pi - 2$$

$$\text{Q8a) } \int \sqrt{\frac{2-x}{2+x}} \, dx \times \frac{\sqrt{2-x}}{\sqrt{2-x}} = \int \frac{2-x}{\sqrt{4-x^2}} \, dx$$

$$= \int \frac{2}{\sqrt{4-x^2}} \, dx - \int \frac{x}{\sqrt{4-x^2}} \, dx$$

$$= 2 \int \frac{1}{\sqrt{4-x^2}} \, dx + \frac{1}{2} \int \frac{-2x}{\sqrt{4-x^2}} \, dx$$

$$\text{Let } u = 4 - x^2 \quad du = -2x \, dx$$

$$\therefore \frac{1}{2} \int \frac{-2x}{\sqrt{4-x^2}} \, dx = \frac{1}{2} \int \frac{1}{\sqrt{u}} \, du = \frac{1}{2} [2\sqrt{u}]$$

$$\therefore \int \sqrt{\frac{2-x}{2+x}} \, dx = 2 \sin^{-1} \frac{x}{2} + \sqrt{4-x^2} + C$$

$$\text{Q8b) i) } \underline{u} \cdot \underline{v} = (3 \times -7) + (-2 \times 6) + (1 \times -5)$$

$$= -38$$

$$\text{Q8b) ii) } \text{proj}_{\underline{v}} \underline{u} = \frac{\underline{u} \cdot \underline{v}}{\underline{v} \cdot \underline{v}} \underline{v}$$

$$= \frac{-38}{(-7)^2 + 6^2 + (-5)^2} \underline{v}$$

$$= \frac{-19}{55} \underline{v}$$

$$\therefore \text{proj}_{\underline{v}} \underline{u} = \frac{-19}{55} \times \begin{bmatrix} -7 \\ 6 \\ -5 \end{bmatrix} = -\frac{133}{55} \begin{bmatrix} 7 \\ 6 \\ 5 \end{bmatrix}$$

$$\text{Q8c) } \int_0^{\frac{\pi}{2}} \frac{1}{4 \sin \theta - 2 \cos \theta + 6} \, d\theta$$

$$t = \tan \frac{\theta}{2} \quad \theta = 0, t = 0 \quad ; \quad \theta = \frac{\pi}{2}, t = 1$$

$$= \int_0^1 \frac{1}{\frac{8t}{1+t^2} - \frac{2(1-t^2)}{1+t^2} + 6} \times \frac{2}{1+t^2} \, dt$$

$$= \int_0^1 \frac{1}{4t-1+t^2+3+3t^2} \, dt = \int_0^1 \frac{1}{4t^2+4t+2} \, dt$$

$$= \int_0^1 \frac{1}{(4t^2+4t+1)+1} \, dt = \frac{1}{2} \int_0^1 \frac{2}{(2t+1)^2+1} \, dt$$

$$= \frac{1}{2} [\tan^{-1}(2t+1)]_0^1 = \frac{1}{2} (\tan^{-1}(3) - \frac{\pi}{4})$$

$$\text{Q9a) } \underline{a} \cdot \underline{b} = (1 \times x) + (2 \times 2) + (x \times 1) = 2x + 4$$

$$\underline{a} \cdot \underline{b} = \sqrt{1^2 + 2^2 + x^2} \sqrt{x^2 + 2^2 + 1^2} \times \cos 60^\circ$$

$$\therefore 2x + 4 = \sqrt{x^2 + 5} \times \sqrt{x^2 + 5} \times \frac{1}{2}$$

$$\therefore 4x + 8 = x^2 + 5 \quad \therefore x^2 - 4x - 3 = 0$$

$$\therefore x = \frac{4 \pm \sqrt{4^2 - 4 \times 1 \times (-3)}}{2} = \frac{4 \pm \sqrt{28}}{2} = 2 \pm \sqrt{7}$$

$$\begin{aligned} \text{Q9b)i) } \int_n^{2n} \frac{dx}{\sqrt{x}} &= [2\sqrt{x}]_n^{2n} \\ &= 2\sqrt{2n} - 2\sqrt{n} \\ &= 2\sqrt{n}(\sqrt{2} - 1) \end{aligned}$$

$$\text{Q9b) ii) } \text{Area of the lower rectangles} < \text{Area under the curve}$$

$$\frac{1}{\sqrt{n+1}} + \frac{1}{\sqrt{n+2}} + \dots + \frac{1}{\sqrt{n}} < \int_n^{2n} \frac{dx}{\sqrt{x}} \\ S_n < 2\sqrt{n}(\sqrt{2} - 1)$$

$$\text{Area under the curve} < \text{Area of the upper rectangles}$$

$$\int_n^{2n} \frac{dx}{\sqrt{x}} < \frac{1}{\sqrt{n}} + \frac{1}{\sqrt{n+1}} + \frac{1}{\sqrt{n+2}} + \dots + \frac{1}{\sqrt{2n-1}}$$

$$2\sqrt{n}(\sqrt{2} - 1) < \frac{1}{\sqrt{n}} + \left(\frac{1}{\sqrt{n+1}} + \frac{1}{\sqrt{n+2}} + \dots + \frac{1}{\sqrt{2n-1}} + \frac{1}{\sqrt{2n}} \right) - \frac{1}{\sqrt{2n}}$$

$$2\sqrt{n}(\sqrt{2} - 1) - \frac{1}{\sqrt{n}} + \frac{1}{\sqrt{2n}} < S_n \quad 2\sqrt{n}(\sqrt{2} - 1) + \frac{1 - \sqrt{2}}{\sqrt{2n}} < S_n$$

$$\therefore 2\sqrt{n}(\sqrt{2} - 1) + \frac{1 - \sqrt{2}}{\sqrt{2n}} < S_n < 2\sqrt{n}(\sqrt{2} - 1)$$

$$\text{Q9b) iii)}$$

$$\text{Let } n = 1\,000\,000$$

$$2\sqrt{1\,000\,000}(\sqrt{2} - 1) + \frac{1 - \sqrt{2}}{\sqrt{2\,000\,000}} < S_{1\,000\,000} < 2\sqrt{1\,000\,000}(\sqrt{2} - 1) \\ 828.4268 < S_{1\,000\,000} < 828.4271$$

$$\therefore S_{1\,000\,000} = 828.427 \quad (\text{to 3 decimal places})$$

$$\frac{1}{\sqrt{1\,000\,000}} + \frac{1}{\sqrt{1\,000\,001}} + \dots + \frac{1}{\sqrt{2\,000\,000}} = \frac{1}{\sqrt{1\,000\,000}} + S_{1\,000\,000} \\ = 828.428 \quad (\text{to 3 decimal places})$$